



# Advanced Time Series Analysis and Forecasting

## Chapter 16: Explosive roots and bubbles



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## Acronyms used in this material (1/2)

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ AUC: Area Under the (ROC) Curve
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BSADF: Backward Supremum Augmented Dickey–Fuller (statistic)
- ▣ BUX: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ CLT: Central Limit Theorem
- ▣ CPI: Consumer Price Index
- ▣ CUSUM: CUMulative SUM
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ FCLT: Functional Central Limit Theorem
- ▣ FRED: Federal Reserve Economic Data
- ▣ FW: Family-Wise (error control)
- ▣ FWER: Family-Wise Error Rate
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GSADF: Generalised Supremum Augmented Dickey–Fuller (test)
- ▣ HICP: Harmonised Index of Consumer Prices
- ▣ LLM: Large Language Model
- ▣ LPPL: Log-Periodic Power Law
- ▣ LPPLS: Log-Periodic Power Law Singularity (model)
- ▣ MC: Monte Carlo
- ▣ MFM: Modelling Financial Markets
- ▣ OLS: Ordinary Least Squares
- ▣ PSY: Phillips, Shi and Yu (2015)
- ▣ PWY: Phillips, Wu and Yu (2011)

## Acronyms used in this material (2/2)

- ▣ PX: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- ▣ ROC: Receiver Operating Characteristic (curve)
- ▣ SADF: Supremum Augmented Dickey–Fuller (test)
- ▣ TSA: Time Series Analysis
- ▣ US: United States
- ▣ WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

## The question of the chapter and the route

- **Question:** when does a price series contain an explosive component, how precisely can we date it, and how much is an alarm worth once false alarms and base rates are counted?
  - ▶ three layers: economic theory of rational bubbles, asymptotic theory of explosive autoregressions, statistical evaluation of detectors
- **Route**
  - ▶ rational bubbles: present value, Blanchard–Watson, the Diba–Grossman critique, Evans bubbles
  - ▶ explosive autoregressions: Cauchy limits, mildly explosive roots; SADF, GSADF, BSADF and their size
  - ▶ date-stamping, real-time monitoring, evaluation of early warnings; fundamentals against prices (rents)
  - ▶ LPPLS as a competing detector; applications: dot-com, Shanghai 2015, Bitcoin, BET 2007, housing
- **Prerequisites and continuation**
  - ▶ [TSA, Chapter 13](#): bubbles in history, PSY basics, LPPL with the Filimonov–Sornette calibration
  - ▶ [Chapter 2](#): unit roots, structural breaks; [MFM, Chapter 17](#) applies the tests to markets

## Self-study guide

- This chapter is for **self-study**: there is no seminar; every section ends with a recap
  - ▶ redo each worked derivation on paper before reading the next slide
  - ▶ after each chart, write your own reading of it, then compare it with the interpretation slide
- Run the lecture notebook: it reproduces every chart and number with smaller Monte Carlo sizes
  - ▶ change one design choice (the minimum window, the lag order, the critical value, the crash threshold) and record how the conclusion changes
- The self-assessment at the end has answers; the quiz of this chapter on the course site is graded
- Prerequisites: Dickey–Fuller asymptotics (Hamilton, Chapter 17), the functional central limit theorem, bootstrap for dependent data ([Chapter 0](#))

## Learning outcomes

- ▣ Derive the general solution of the present-value model and state the restrictions of Diba and Grossman on rational bubbles
- ▣ Derive the Cauchy limit of the OLS estimator under an explosive root and explain why the mildly explosive limit is invariant to the error law
- ▣ Compute SADF, GSADF and BSADF with Monte Carlo and wild-bootstrap critical values, and control false alarms across dates
- ▣ Measure date-stamping delays and evaluate early-warning signals with hit rates, false-alarm rates, precision and ROC curves
- ▣ Separate explosive prices from explosive fundamentals, and compare PSY with LPPLS confidence indicators on real episodes

## Reading, data and tools

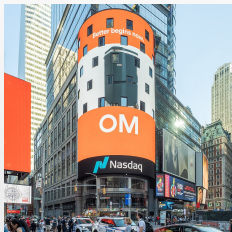
- Theory: Blanchard and Watson (1982); Diba and Grossman (1988a); Evans (1991); Phillips and Magdalinos (2007); Phillips, Shi and Yu (2015a); Phillips, Shi and Yu (2015b)
  - ▶ Inference and monitoring: Phillips and Shi (2020); Harvey et al. (2016); Homm and Breitung (2012); Astill et al. (2018); survey: Gürkaynak (2008)
  - ▶ LPPLS: Johansen, Ledoit and Sornette (2000); Filimonov and Sornette (2013); Filimonov, Demos and Sornette (2017); Shu and Zhu (2020); critique: Feigenbaum (2001); Geraskin and Fantazzini (2013)
- Python Quantlets of this chapter: [Quantlets/Ch\\_16](#)
  - ▶ every window regression written out in `numpy` (cumulated cross-products); the R package `exuber` Vasilopoulos, Pavlidis and Martínez-García (2022) implements the same statistics
- Lecture notebook: [open in Google Colab](#)

## Data used in this chapter

| Series                                                | Source                                                                                                    | Sample            |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------|
| S&P 500, Nasdaq 100, Shanghai Composite, BET, Bitcoin | EODHD daily closes (weekly: Friday close)                                                                 | 1990 – 2026       |
| US house prices, rents, consumer prices               | FRED: S&P CoreLogic Case–Shiller national index, CPI rent of primary residence, CPI (seasonally adjusted) | 1987 – July 2026  |
| Romanian house prices, rents, consumer prices         | Eurostat: house price index; HICP actual rentals and all items                                            | 2009 – March 2026 |
| All price series of the course data                   | EODHD, monthly closes (screen of the AI mini-case)                                                        | 1990 – 2026       |

- Tests run on log prices or log ratios; weekly data for stock indices and Bitcoin, monthly or quarterly data for housing

## Known from TSA and new here



Nasdaq MarketSite, Times Square, 2021

Photo: [ajay\\_suresh](#) (2021); CC BY 2.0; Wikimedia

[Commons](#)

- Known (TSA, Chapter 13)
  - ▶ manias in history; the right-tailed ADF, SADF, GSADF and BSADF with Monte Carlo critical values
  - ▶ the LPPL equation, the two-step calibration, the confidence indicator
- New: the research layer
  - ▶ why rational bubbles must explode and why they are hard to see
  - ▶ the limit theory behind the tests, their size under changing volatility and across many dates
  - ▶ how early, how often and how wrongly the detectors raise alarms
- Case studies, on our data
  - ▶ Phillips, Wu and Yu (2011); Phillips, Shi and Yu (2015a); Phillips and Yu (2011)
  - ▶ Homm and Breitung (2012); Harvey et al. (2016); Sornette et al. (2015)

## The present-value model and its bubble solutions (1/2)

- **No-arbitrage** with a constant required return  $r > 0$ : today's price is the discounted expected value of tomorrow's price plus dividend

$$P_t = (1 + r)^{-1} \mathbb{E}_t[P_{t+1} + D_{t+1}]$$

►  $P_t$ : price;  $D_t$ : dividend (or rent);  $\mathbb{E}_t$ : expectation given the information available at  $t$

- **Forward substitution**  $k$  times: the price is the discounted dividends of the next  $k$  periods plus the discounted price at  $t + k$

$$P_t = \sum_{i=1}^k (1 + r)^{-i} \mathbb{E}_t D_{t+i} + (1 + r)^{-k} \mathbb{E}_t P_{t+k}$$

- If the **transversality condition**  $\lim_{k \rightarrow \infty} (1 + r)^{-k} \mathbb{E}_t P_{t+k} = 0$  holds, the price equals the **fundamental value**

$$P_t = F_t := \sum_{i \geq 1} (1 + r)^{-i} \mathbb{E}_t D_{t+i}$$

## The present-value model and its bubble solutions (2/2)

- **General solution:** the fundamental plus a **bubble**  $B_t$  whose expected value grows at the rate  $r$

$$P_t = F_t + B_t, \quad \mathbb{E}_t B_{t+1} = (1+r)B_t, \quad \mathbb{E}_t B_{t+k} = (1+r)^k B_t$$

- ▶  $B_t$  is a submartingale with an explosive mean: it satisfies the no-arbitrage equation but violates transversality
- **Econometric content:**  $B_t$  follows  $(1 - (1+r)L)B_t = z_t$ , an AR(1) with coefficient  $1+r > 1$ 
  - ▶  $L$ : the lag operator ( $LB_t = B_{t-1}$ );  $z_t$ : a martingale difference ( $\mathbb{E}_{t-1} z_t = 0$ ); an **explosive autoregressive root**
- General equilibrium limits [Tirole \(1985\)](#); [Santos and Woodford \(1997\)](#)
  - ▶ with infinitely lived agents bubbles are ruled out in most economies
  - ▶ with overlapping generations they can exist if growth exceeds  $r$

## Worked example: the fundamental value with random-walk dividends

- Let dividends be a random walk with drift,  $D_t = \mu + D_{t-1} + \varepsilon_t$ : then  $\mathbb{E}_t D_{t+i} = D_t + i\mu$ 
  - $\mu$ : the expected change per period (drift);  $\varepsilon_t$ : a zero-mean shock

- Summing the discounted expected dividends

$$F_t = D_t \sum_{i \geq 1} (1+r)^{-i} + \mu \sum_{i \geq 1} i(1+r)^{-i} = \frac{D_t}{r} + \frac{\mu(1+r)}{r^2}$$

- using  $\sum_{i \geq 1} x^i = x/(1-x)$  and  $\sum_{i \geq 1} ix^i = x/(1-x)^2$  with  $x = (1+r)^{-1}$

- Implications:  $F_t$  is  $I(1)$  like  $D_t$ ;  $P_t - D_t/r$  is stationary without a bubble: price and dividend are **cointegrated** with vector  $(1, -1/r)$  [Campbell and Shiller \(1987\)](#)
  - with a bubble,  $P_t - D_t/r - \mu(1+r)/r^2 = B_t$  inherits the explosive root: no difference  $\Delta^d P_t$  is stationary [Diba and Grossman \(1988b\)](#)
- Check on paper:  $r = 0.05$ ,  $\mu = 0.04$ ,  $D_t = 2$  give  $F_t = 40 + 16.8 = 56.8$ ; the price-dividend ratio 28.4 is constant only if  $\mu = 0$

## Blanchard–Watson: a bubble that can burst



Olivier Blanchard, IMF, 2008

Photo: Eugene Salazar, IMF (2008); public domain; Wikimedia Commons

- Blanchard and Watson (1982): each period the bubble survives with probability  $\pi$  and bursts with probability  $1 - \pi$

$$B_{t+1} = \begin{cases} \frac{1+r}{\pi} B_t + \varepsilon_{t+1} & \text{with probability } \pi \\ \varepsilon_{t+1} & \text{with probability } 1 - \pi \end{cases}$$

►  $\varepsilon_{t+1}$ : a zero-mean shock;  $\pi \in (0, 1)$ : the survival probability

- Check:  $\mathbb{E}_t B_{t+1} = \pi \cdot \frac{1+r}{\pi} B_t = (1+r)B_t$ : a valid rational bubble
- Survival time is geometric: expected duration  $1/(1 - \pi)$ ;  $\pi = 0.98$  gives 50 periods
- While it survives it grows at  $(1+r)/\pi - 1 > r$ : the extra return pays for the crash risk
- After a burst it restarts from noise, which can be negative: the next slide shows why that is a problem

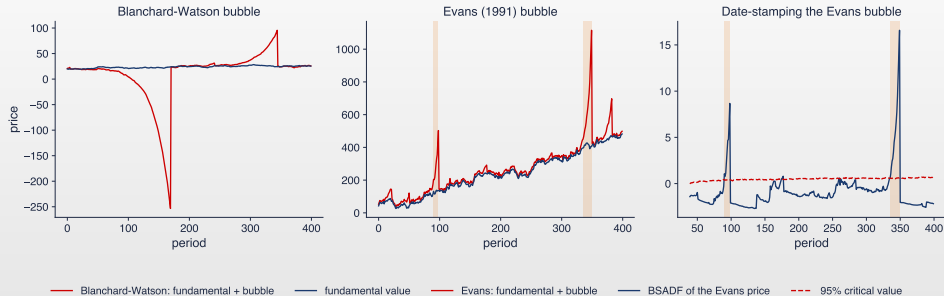
## The Diba–Grossman critique

- Write  $B_{t+1} = (1 + r)B_t + z_{t+1}$  with  $\mathbb{E}_t z_{t+1} = 0$  Diba and Grossman (1988a)
- **No negative bubbles:** if  $B_t < 0$ , then  $\mathbb{E}_t B_{t+k} = (1 + r)^k B_t \rightarrow -\infty$ , so the expected price eventually becomes negative; free disposal ( $P \geq 0$ ) rules this out
- **No birth:** if  $B_t = 0$ , then  $\mathbb{E}_t B_{t+1} = 0$  and  $B_{t+1} \geq 0$ , hence  $B_{t+1} = 0$  almost surely; a bubble that exists today existed on the first trading day
  - ▶ the Blanchard–Watson restart from noise violates both points; a bubble can still burst, but not restart
- **Test Diba and Grossman (1988b):** no bubble  $\Rightarrow P_t$  and  $D_t$  cointegrated and  $\Delta P_t$  stationary; on annual US data they found no explosive component
  - ▶ unobserved fundamentals or a time-varying  $r$  make the test a joint test of the bubble and of the pricing model Gürkaynak (2008)

## Evans: periodically collapsing bubbles

- Parameters  $0 < \delta < (1+r)\alpha$ ,  $u_{t+1} > 0$  i.i.d. with  $\mathbb{E}u = 1$ ,  $\theta_{t+1} \sim \text{Bernoulli}(\pi)$  [Evans \(1991\)](#)
  - ▶  $B_{t+1} = (1+r)B_t u_{t+1}$  if  $B_t \leq \alpha$  (slow growth phase)
  - ▶  $B_{t+1} = [\delta + \pi^{-1}(1+r)\theta_{t+1}(B_t - \delta/(1+r))] u_{t+1}$  if  $B_t > \alpha$  (fast phase that collapses to  $\delta$ )
- Exercise: show  $\mathbb{E}_t B_{t+1} = (1+r)B_t$  in both regimes (use  $\mathbb{E}\theta = \pi$ ); the bubble is positive and never dies, consistent with Diba–Grossman
- Evans' point: on the whole sample, frequent collapses make  $P_t$  look like an  $I(1)$  or even stationary process; unit-root and cointegration tests have almost no power
  - ▶ the full-sample OLS slope pairs the largest falls with the highest lagged levels: collapses pull  $\hat{\delta}$  down
- The remedy of [Phillips, Wu and Yu \(2011\)](#) and [Phillips, Shi and Yu \(2015a\)](#): estimate on sub-samples (windows) and take the supremum, so that some windows contain the expansion but not the collapse

## Simulated rational bubbles and their detection



- Left: Blanchard–Watson,  $r = 0.02$ ,  $\pi = 0.98$ ; middle: Evans bubble times 20 on the fundamental of the worked example ( $r = 0.05$ ,  $\alpha = 1$ ,  $\delta = 0.5$ ,  $\pi = 0.85$ ); right: BSADF of the Evans price against its pointwise 95% critical value (2 000 Monte Carlo paths)
- Whole-sample ADF  $-2.81$  (95% critical value  $-0.07$ ); SADF 6.80 (1.45); GSADF 16.58 (2.18); 2 dated episodes

## Interpreting the simulated bubbles

- The Blanchard–Watson path restarts below zero after a burst and then explodes downwards: exactly the negative bubble that Diba and Grossman exclude
- The Evans price looks like its fundamental most of the time; the whole-sample ADF ( $-2.81$ ) is even below its 95% critical value: Evans' pitfall in one number
- The window statistics see the large expansions: GSADF 16.58 against 2.18; the BSADF sequence crosses its critical value only in the long runs
- The minimum-duration rule (6 consecutive exceedances) keeps 2 episodes and drops the short bursts: power is traded for fewer false alarms

## Recap: rational bubbles

- Price = fundamental + bubble, with  $\mathbb{E}_t B_{t+1} = (1 + r)B_t$ : a bubble is an explosive autoregressive component
- Diba–Grossman: bubbles cannot be negative and cannot start later; they can only grow or burst
- Evans: periodic collapses hide the bubble from whole-sample tests; recursive window tests restore the power
- Every bubble test is a joint test with the model of the fundamental (a constant  $r$ , observed dividends or rents)

## Three regimes of the AR(1) estimator (1/2)

| Root                                                   | Rate and limit of $\hat{\rho} - \rho$                                    | Invariance                                         |
|--------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------|
| $ \rho  < 1$                                           | $\sqrt{n}(\hat{\rho} - \rho) \Rightarrow N(0, 1 - \rho^2)$               | yes (CLT)                                          |
| $\rho = 1$                                             | $n(\hat{\rho} - 1) \Rightarrow \int_0^1 W dW / \int_0^1 W^2$             | yes (FCLT)                                         |
| $\rho = 1 + c/k_n, k_n \rightarrow \infty, k_n = o(n)$ | $\frac{k_n \rho_n^n}{2c_n}(\hat{\rho} - \rho_n) \Rightarrow \mathcal{C}$ | yes Phillips and Magdalinos (2007)                 |
| $ \rho  > 1$ fixed                                     | $\frac{\rho^n}{\rho^2 - 1}(\hat{\rho} - \rho) \Rightarrow \mathcal{C}$   | only Gaussian errors White (1958); Anderson (1959) |

□ Model:  $y_t = \rho y_{t-1} + u_t$ ,  $y_0 = 0$ ,  $u_t$  i.i.d.  $(0, \sigma^2)$ ;  $\hat{\rho}$ : OLS without intercept

- $n$ : sample size;  $W$ : a standard Brownian motion;  $\mathcal{C}$ : the standard Cauchy law;  $\Rightarrow$ : convergence in distribution
- CLT/FCLT: (functional) central limit theorem

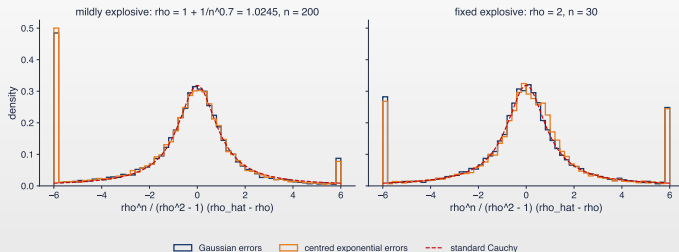
## Three regimes of the AR(1) estimator (2/2)

- Mildly explosive root:  $\rho_n = 1 + c/k_n$ 
  - ▶  $c > 0$ , and  $k_n \rightarrow \infty$  more slowly than  $n$ :  $k_n = o(n)$ , i.e.  $k_n/n \rightarrow 0$
  - ▶ the Cauchy limit is derived in the [Appendix](#)
- The rate grows from  $\sqrt{n}$  to  $n$  to the exponential  $\rho^n$ 
  - ▶ the further from unity, the faster we learn  $\rho$
  - ▶ unit roots: [Chapter 2](#) and [Hamilton \(1994\)](#), Chapter 17
- Mildly explosive roots are the realistic case for bubbles
  - ▶ faster than any local-to-unity root  $1 + c/n$
  - ▶ slower than any fixed explosive root
  - ▶ their limit theory is invariant to the error law, unlike the fixed explosive case

## Worked derivation: where the Cauchy law comes from

- $\hat{\rho} - \rho = \sum_t y_{t-1} u_t / \sum_t y_{t-1}^2$  and  $y_t = \sum_{j \leq t} \rho^{t-j} u_j$
- Define  $Y_n = \sum_{j=1}^n \rho^{-j} u_j$  (dominated by the **first** shocks) and  $X_n = \sum_{j=1}^n \rho^{-(n-j)-1} u_j$  (dominated by the **last** shocks)
  - ▶ then  $\rho^{-n} y_n \approx Y_n$ ,  $\rho^{-2n} \sum_t y_{t-1}^2 \approx Y_n^2 / (\rho^2 - 1)$  and  $\rho^{-n} \sum_t y_{t-1} u_t \approx X_n Y_n$
- Hence  $\frac{\rho^n}{\rho^2 - 1} (\hat{\rho} - \rho) \approx \frac{X_n}{Y_n}$ 
  - ▶  $X_n$  and  $Y_n$  use disjoint blocks of shocks asymptotically: independent, with equal variances  $\sigma^2 / (\rho^2 - 1)$
  - ▶ **fixed**  $\rho$ : each sum is dominated by a few shocks; it is Gaussian only if the  $u_j$  are Gaussian, so the ratio is Cauchy only then [Anderson \(1959\)](#)
  - ▶ **mildly explosive**  $\rho_n$ : about  $k_n \rightarrow \infty$  shocks enter each sum with comparable weights, a CLT applies, and  $X_n / Y_n \Rightarrow C$  for any i.i.d. errors with finite variance [Phillips and Magdalinos \(2007\)](#)
- Pivotal inference: the 95% interval  $\hat{\rho} \pm (\hat{\rho}^2 - 1) \hat{\rho}^{-n} \tan(0.475\pi)$ , with  $\tan(0.475\pi) \approx 12.71$ , replaces  $\rho$  by  $\hat{\rho}$  in the norming factor

## Cauchy limits in finite samples



- 20 000 samples; normalised error  $\rho^n(\hat{\rho} - \rho)/(\rho^2 - 1)$  with Gaussian and centred exponential errors (skewed), against the Cauchy density
- Kolmogorov–Smirnov distance to Cauchy: mild,  $n = 200$ : 0.060 and 0.062;  $n = 2000$  ( $\rho = 1.0049$ ): 0.011 ( $p = 0.620$ ) and 0.007 ( $p = 0.945$ ); fixed  $\rho = 2$ : 0.007 ( $p = 0.263$ ) and 0.025 ( $p < 0.001$ )

## Interpreting the Cauchy limits

- ▣ Fixed root: with Gaussian errors the Cauchy law fits ( $p = 0.263$ ); with skewed errors it is rejected ( $p < 0.001$ ): no invariance principle, as Anderson showed
- ▣ Mildly explosive root: at  $n = 2000$  both error laws give the Cauchy law; at  $n = 200$  ( $k_n = 40.8$ ) the approximation is still rough for both: the CLT inside  $X_n$  and  $Y_n$  needs many effective shocks
- ▣ Coverage of the 95% Cauchy interval: 97.0% and 97.2% ( $n = 200$ ), 95.0% and 94.7% ( $n = 2000$ ): conservative in small samples, accurate in large ones
- ▣ For bubble tests this is good news: the explosive alternative is learned at an exponential rate, so the window statistics diverge fast once a window sits inside a bubble

## From limit theory to bubble tests

- Under a mildly explosive alternative the ADF  $t$ -statistic diverges to  $+\infty$ : right-tailed tests are **consistent** Phillips, Shi and Yu (2015b)
- Levels against logs: a rational bubble is explosive in the **level**; in logs a pure bubble path  $\ln B_0 + t \ln(1 + r)$  is a linear trend
  - ▶  $\ln(F_t + B_t)$  **accelerates** while the bubble share rises: log-price tests target accelerating log prices, which is what PWY and PSY test in practice
- Specification matters: the intercept, a weak drift  $dT^{-\eta}$  ( $\eta > 1/2$ ) in the null, and the lag order change the null limit and the size Phillips, Shi and Yu (2014)
- Collapses: a sudden fall is a mildly **integrated** or stationary phase after the explosive one; Phillips and Shi (2018) date the implosion with a reverse regression

## Recap: explosive autoregressions

- Explosive roots are learned at rate  $\rho^n$ ; the limit is Cauchy, a ratio of two independent normals built from the first and the last shocks
- Fixed explosive roots have no invariance principle; mildly explosive roots do, which makes them the working model for bubbles
- The Cauchy interval for  $\rho$  is pivotal but needs large samples; right-tailed tests are consistent against mildly explosive alternatives

## The recursive statistics (1/2)

- **Window regression** on the log price  $y_t$ , over the sample fractions  $[r_1, r_2]$

$$\Delta y_t = a + \delta y_{t-1} + \sum_{j=1}^k \phi_j \Delta y_{t-j} + e_t, \quad t = \lfloor r_1 T \rfloor, \dots, \lfloor r_2 T \rfloor$$

- ▶  $\Delta y_t = y_t - y_{t-1}$ ;  $a$ : intercept;  $\delta$ : the coefficient tested;  $\phi_j$ : coefficients of  $k$  lagged differences;  $e_t$ : error;  $T$ : sample size;  $\lfloor \cdot \rfloor$ : integer part
- ▶  $\text{ADF}_{r_1}^{r_2}$ : the  $t$ -statistic of  $\delta$  on this window
- $H_0$ :  $\delta = 0$  (unit root) against  $H_1$ :  $\delta > 0$  (explosive): a **right-tailed** test
  - ▶ minimum window fraction  $r_0 = 0.01 + 1.8/\sqrt{T}$
- **Suprema** over windows Phillips, Wu and Yu (2011), Phillips, Shi and Yu (2015a)

$$\text{SADF} = \sup_{r_2 \in [r_0, 1]} \text{ADF}_{r_1}^{r_2}, \quad \text{BSADF}_{r_2} = \sup_{r_1 \in [0, r_2 - r_0]} \text{ADF}_{r_1}^{r_2}, \quad \text{GSADF} = \sup_{r_2} \text{BSADF}_{r_2}$$

- ▶ SADF: windows starting at the first observation; BSADF: all windows ending at  $r_2$  (used to date); GSADF: all windows

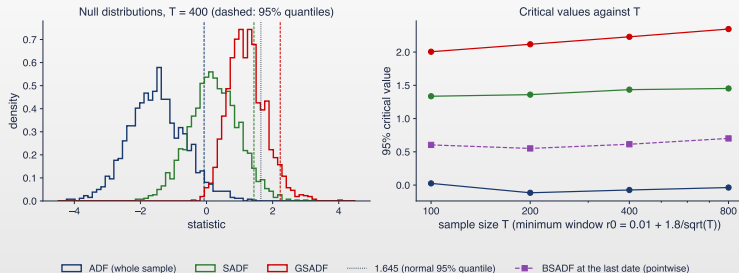
## The recursive statistics (2/2)

- Null limit (PSY, Theorem 1), with  $r_w = r_2 - r_1$  and integrals over  $[r_1, r_2]$ :

$$\text{ADF}_{r_1}^{r_2} \Rightarrow \frac{\frac{1}{2} r_w [W(r_2)^2 - W(r_1)^2 - r_w] - \int W dr [W(r_2) - W(r_1)]}{r_w^{1/2} \left\{ r_w \int W^2 dr - \left( \int W dr \right)^2 \right\}^{1/2}}$$

- ▶  $W$ : a standard Brownian motion;  $\Rightarrow$ : convergence in distribution as  $T \rightarrow \infty$
- ▶ pivotal: free of  $\sigma$  and of the drift, so critical values depend only on  $r_0$ , the deterministic terms and  $T$  in finite samples
- Computation: for  $T = 400$  there are 65 341 window regressions; cumulated cross-products give all of them in  $O(T^2)$  operations

## Null distributions and critical values



- Null model  $y_t = T^{-1} + y_{t-1} + e_t$ ,  $e_t \sim N(0, 1)$ , no lags; 95% critical values at  $T = 100, 200, 400, 800$ : SADF 1.34, 1.36, 1.43, 1.45; GSADF 2.01, 2.12, 2.23, 2.35

## Interpreting the null distributions

- ▣ The whole-sample ADF has a negative 95% quantile ( $-0.07$  at  $T = 400$ ): using  $1.645$  would almost never reject, the right tail of the Dickey–Fuller law is short
- ▣ Each supremum shifts the distribution to the right: SADF searches over end points, GSADF over start and end points; the GSADF bar is the price of the search
- ▣ The GSADF critical value rises with  $T$  ( $2.01$  to  $2.35$ ) although  $r_0$  falls: more windows, a larger maximum; tabulate critical values for your own  $T$ ,  $r_0$  and lag order
- ▣ The pointwise 95% quantile of BSADF at the last date ( $0.61$  at  $T = 400$ ) is far below the GSADF one: a single date is tested at 5%, the whole path is not

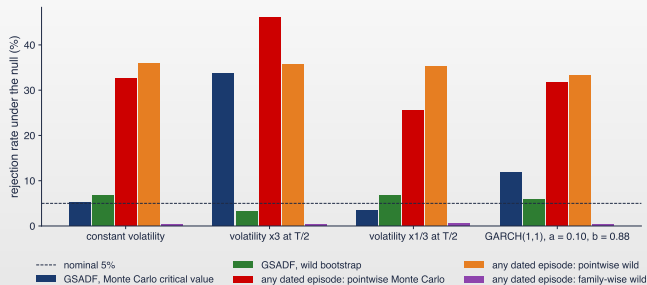
## Changing volatility and the wild bootstrap

- The pivotal limit assumes homoskedastic shocks; with a volatility shift the limit involves the variance profile, and Monte Carlo critical values give the wrong size [Harvey et al. \(2016\)](#)
  - ▶ a late rise in volatility looks like acceleration; a fall in volatility makes the test conservative
- **Wild bootstrap** [Phillips and Shi \(2020\)](#): fit the null  $\Delta y_t = \mu + \sum_j \phi_j \Delta y_{t-j} + e_t$ , draw Rademacher signs  $w_t = \pm 1$ , build  $\Delta y_t^* = \sum_j \hat{\phi}_j \Delta y_{t-j}^* + w_t \hat{e}_t$ , cumulate, recompute the statistics
  - ▶ the bootstrap series has a unit root by construction and the **same volatility pattern**  $|\hat{e}_t|$  as the data
  - ▶ critical values: pointwise quantiles of BSADF, the quantile of the maximum (GSADF), or the quantile of the maximum over a moving window
- Alternatives for non-stationary volatility: weighted and sign-based statistics [Harvey, Leybourne and Zu \(2020\)](#); end-of-sample tests [Astill et al. \(2017\)](#)

## Many dates, one false alarm

- Date-stamping compares  $\text{BSADF}_{r_2}$  with a **pointwise** critical value at every date: each date has a 5% false-alarm chance, but a sample has hundreds of dates
  - ▶ BSADF is highly autocorrelated in  $r_2$ , so false alarms come in runs: the minimum-duration rule removes the shortest runs only
- **Family-wise error rate** (FWER): the probability of at least one false alarm over a set of dates
  - ▶ over the whole sample: compare BSADF with the 95% quantile of its maximum, i.e. the GSADF critical value
  - ▶ over a moving window of  $\tau_b$  dates [Phillips and Shi \(2020\)](#): the 95% bootstrap quantile of  $\max_{s \in (t - \tau_b, t]} \text{BSADF}_s$ ; here  $\tau_b = 52$  weeks
- Asymptotically, consistent dating needs critical values that diverge slowly, so that the false-alarm probability vanishes [Phillips, Shi and Yu \(2015b\)](#); in practice a fixed 5% at every date is used, and its cost must be measured

## Size under changing volatility and across dates



- 400 null paths of  $T = 200$  per design, each with its own wild bootstrap (199 draws); a dated episode needs 6 consecutive exceedances; nominal level 5%

ATS\_ch16\_size\_wild\_bootstrap

## Interpreting the size study

- GSADF with Monte Carlo critical values: 5.2% with constant volatility, 33.8% after a late tripling, 11.8% under GARCH; the wild bootstrap gives 6.8%, 3.2% and 5.8%
- Pointwise dating finds at least one episode in 32.5% (Monte Carlo) and 36.0% (wild) of paths **without any bubble**, even with constant volatility
- With the family-wise threshold over the whole sample, false episodes fall to 0.2%–0.5%: below 5%, because an episode also needs 6 consecutive exceedances; the cost is power and later dates
- Lesson: report which error rate a dated episode controls; a 5% pointwise test on every week is not a 5% test of the sample

## Recap: recursive tests

- ▣ SADF, GSADF and BSADF are suprema of window ADF statistics with a pivotal null limit; critical values must match  $T$ ,  $r_0$ , the deterministic terms and the lags
- ▣ Changing volatility distorts the size; the wild bootstrap keeps the volatility pattern and restores it
- ▣ Pointwise dating has a high false-episode rate over a sample; family-wise thresholds control it at the cost of power

## Date-stamping rules and their accuracy

- **Origination and termination** Phillips, Shi and Yu (2015a): the first date where BSADF exceeds its critical value, and the first later date where it falls back below

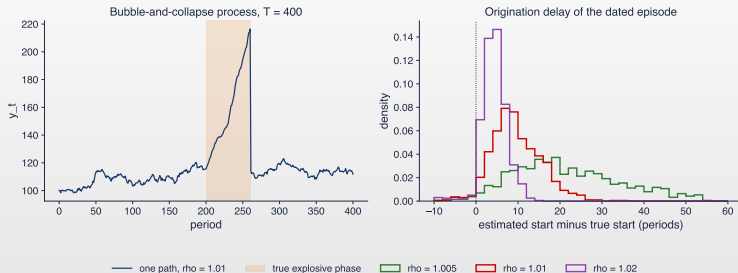
$$\hat{r}_e = \inf\{r_2 \geq r_0 : \text{BSADF}_{r_2} > cv_{r_2}\}, \quad \hat{r}_f = \inf\{r_2 \geq \hat{r}_e + \delta \ln(T)/T : \text{BSADF}_{r_2} < cv_{r_2}\}$$

- ▶  $cv_{r_2}$ : the pointwise 95% critical value at  $r_2$ ;  $\delta \ln(T)/T$ : a minimum duration (with  $\delta$  a tuning constant)
- ▶ here: an episode is a run of at least  $\lceil \ln T \rceil$  consecutive exceedances, dated from the first to the last one (as in exuber Vasilopoulos, Pavlidis and Martínez-García (2022))
- In real time an episode is **confirmed** only after the minimum duration: the alarm date is the start plus  $\lceil \ln T \rceil - 1$
- Consistency of  $(\hat{r}_e, \hat{r}_f)$  holds for mildly explosive bubbles of fixed fractional duration Phillips, Shi and Yu (2015b)
  - ▶ the start is dated late, because a window must first accumulate explosive observations
  - ▶ improved start and end estimators that use the whole bubble period: Harvey, Leybourne and Sollis (2017)

## Date-stamping: design of the experiment

- Simulated paths: random walk from  $y_0 = 100$ , then an explosive phase, a collapse, and a random walk again
  - ▶ explosive phase  $y_t = \rho y_{t-1} + e_t$ ,  $\rho > 1$ , on periods 200–260;  $e_t$ : Normal noise
  - ▶ collapse to the pre-bubble level at the end of the explosive phase
  - ▶  $T = 400$ , 1 000 paths for each value of  $\rho$
- Questions: how often is the bubble found, how late is its start dated, how often is a false episode dated before it?

## How late is the start dated?



- Delay of the first dated exceedance after the true start (periods); pointwise 95% Monte Carlo critical values; growth over the bubble  $\rho^{60}$ : 1.35, 1.82 and 3.28

## Interpreting the date-stamping experiment

- ▣ Detection rates 94.9%, 100.0% and 100.0% for  $\rho = 1.005, 1.01, 1.02$ : a bubble of 60 periods is almost always found
- ▣ Median start delays 19, 8 and 4 periods (interquartile ranges 11–29, 5–13, 2–6); confirmation adds 6 – 1 periods: medians 24, 13, 9
- ▣ For the slowest bubble the alarm comes after about a third of its life: a detector of **ongoing** exuberance, not of its birth
- ▣ In 26%–31% of paths a false episode is dated **before** the bubble starts: the multiplicity of the previous section, now inside a true-bubble design

## Real-time monitoring (1/2)

- Monitoring: a training sample of  $n$  observations without a bubble, then a decision at each new date  $t > n$ ; the error rate is over the **whole** monitoring horizon [Chu, Stinchcombe and White \(1996\)](#)
- CUSUM** of returns: cumulated deviations of the new returns from the training mean, in units of the training volatility

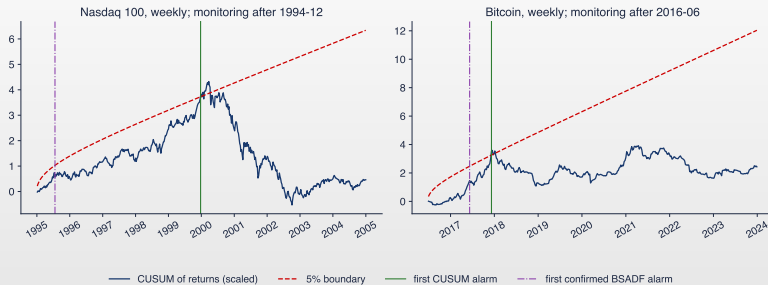
$$S_t = \sum_{j=n+1}^t \frac{r_j - \bar{r}_n}{\hat{\sigma}_n \sqrt{n}}, \quad \text{alarm when } S_t > b_t = \sqrt{\frac{t-n}{n} \cdot \frac{t}{n} \left[ a^2 + \ln \frac{t}{t-n} \right]}$$

- $r_j$ : return of period  $j$ ;  $\bar{r}_n$ ,  $\hat{\sigma}_n$ : mean and standard deviation of the training returns;  $b_t$ : the boundary, which widens with  $t$ ;  $a$ : a constant that sets the error rate
- Error rate over the whole horizon (proof in the [Appendix](#))
  - $\Pr\{S_t > b_t \text{ for some } t\} \rightarrow 1 - \Phi(a) + a\varphi(a)$ ;  $\Phi$ ,  $\varphi$ : standard Normal distribution and density functions
  - 5% gives  $a = 2.50$ ,  $a^2 = 6.25$

## Real-time monitoring (2/2)

- Simulated size,  $n = 100$ , horizon  $5n$ , 2000 paths
  - ▶ 3.5% with i.i.d. returns, 6.0% with GARCH returns
- [Homm and Breitung \(2012\)](#) compare bubble tests and monitoring schemes and recommend monitoring of this kind for real time
  - ▶ [Astill et al. \(2018\)](#) give monitoring statistics with controlled size for explosive bubbles
- BSADF in real time is also a monitor:  $BSADF_t$  uses only data up to  $t$ , but its pointwise critical values do not control the error over the horizon

## Monitoring the Nasdaq 100 and Bitcoin



- Weekly log returns; training: 1990–1994 (Nasdaq 100,  $n = 260$ ) and September 2014 – June 2016 (Bitcoin,  $n = 92$ ); BSADF with pointwise Monte Carlo critical values

## Interpreting the monitoring charts

- ▣ Nasdaq 100: the CUSUM alarm comes on 24 December 1999, three months before the peak of 24 March 2000; the first confirmed BSADF alarm after training comes on 21 July 1995, more than four years earlier
- ▣ Bitcoin: CUSUM on 8 December 2017, at the top of the 2017 run-up; BSADF on 9 June 2017
  - ▶ the boundary widens with  $\sqrt{t \ln t}$ , so after 2018 the 2021 rally cannot reach it
- ▣ CUSUM detects a change in the mean return (persistent acceleration), not explosiveness as such; it is slow, but its size is controlled over the whole horizon
- ▣ BSADF reacts early and often: an early alarm that is followed by years of further gains is costly to anyone who acts on it

## Evaluating early warnings honestly

- ▣ Define the **event** before looking at alarms: here a fall of at least 20% within 182 days after the date; define the **alarm** rule and the evaluation dates (every fifth trading day)
- ▣ Hit rate  $H = \Pr(\text{alarm} \mid \text{event})$ , false-alarm rate  $F = \Pr(\text{alarm} \mid \text{no event})$ , base rate  $\pi_0 = \Pr(\text{event})$ 
  - ▶ **precision** by Bayes:  $\Pr(\text{event} \mid \text{alarm}) = H\pi_0 / [H\pi_0 + F(1 - \pi_0)]$ ; with  $\pi_0 = 5\%$ ,  $H = 80\%$ ,  $F = 20\%$  it is only 17%
  - ▶ the noise-to-signal ratio  $F/H$  of the early-warning literature [Kaminsky, Lizondo and Reinhart \(1998\)](#)
- ▣ **ROC curve**:  $(F, H)$  for every threshold of a continuous score; **AUC** =  $\Pr(\text{score of an event date} > \text{score of a non-event date})$ ; 0.5 means no skill
- ▣ Overlapping horizons make evaluation dates strongly dependent: intervals for the AUC need a moving-block bootstrap ([Chapter 0](#)), and a few crashes carry the whole sample

## Recap: date-stamping and early warning

- BSADF dates the start late (after a run of explosive observations) and confirms it later still; slow bubbles are found after a large part of their life
- Monitoring with a boundary controls the error over the whole horizon; pointwise BSADF does not
- Evaluate alarms with hit rate, false-alarm rate, precision at the base rate and ROC/AUC, with block-bootstrap intervals

## Testing the ratio, the price and the fundamental



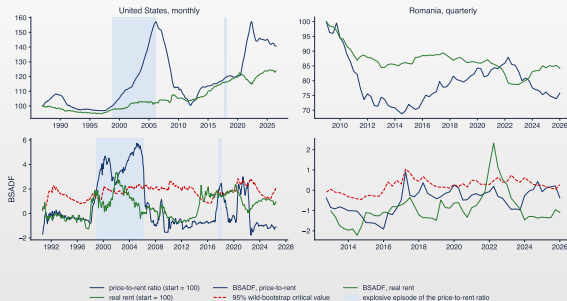
A foreclosed home in the United States, 2008

Photo: Brendel (2008); CC BY-SA 3.0; Wikimedia

[Commons](#)

- Under the present-value model the price-to-rent (or price-dividend) ratio is stationary; an explosive **ratio** with a non-explosive **fundamental** points to a bubble
- If rents themselves explode, an explosive price is not evidence of a bubble: test both (the logic of [Diba and Grossman \(1988b\)](#) applied with PSY)
- Landmarks: [Phillips and Yu \(2011\)](#) date the US price-to-rent exuberance before the subprime crisis; [Pavlidis et al. \(2016\)](#) apply GSADF to the international housing panel of the Dallas Fed
- Monthly house prices are smoothed (repeat-sales indices, appraisals): their changes are autocorrelated, so the lag order  $k$  matters; here  $k$  by BIC, up to 6 months

## US and Romanian housing: ratio and fundamental



- Log price-to-rent and log real rent; BSADF with  $k$  lags (US ratio  $k = 1$ , rent  $k = 1$ ) and pointwise 95% wild-bootstrap critical values (999 draws); shaded: dated episodes of the ratio

## Interpreting the housing tests

- US price-to-rent: GSADF 5.75 against the wild-bootstrap value 4.47; the main episode runs from November 1998 to March 2006, the boom before the subprime crisis found by [Phillips and Yu \(2011\)](#)
- US real rent: GSADF 3.35 against 3.25, with episodes February 1998–March 1999; June 2001–September 2004; September 2014–May 2021; September 2023–April 2026: the fundamental also accelerated, so the 2014–2022 price rise is partly a rent story
- US real house price alone: episodes February 2002–March 2006; December 2020–May 2022; the 2020–2022 surge is explosive in the price but not in the ratio, unlike 1998–2006
- Romania (quarterly,  $T = 68$ ): GSADF of the ratio 0.91 against 1.38, of the real price 0.88 against 2.11: no explosive episode since 2009; the HICP rent index jumps by 33% in April 2026, so the sample ends in March 2026

## Recap: fundamentals against prices

- Test the ratio and the fundamental separately: an explosive ratio with a calm fundamental is the bubble signature
- Smoothed housing indices need lag augmentation and a bootstrap under the AR null; without lags the tests reject almost everywhere
- Short samples (Romania since 2009) have little power: no rejection is not evidence of no bubble

## LPPLS in one slide



Didier Sornette, 2012

Photo: [Didier Sornette \(2012\)](#); CC BY-SA  
3.0 de; [Wikimedia Commons](#)

- [Johansen, Ledoit and Sornette \(2000\)](#): the log price grows faster than exponentially towards a critical time  $t_c$ , with oscillations that speed up
$$\ln p(t) = A + B(t_c - t)^m + C_1(t_c - t)^m \cos(\omega \ln(t_c - t)) + C_2(t_c - t)^m \sin(\omega \ln(t_c - t))$$
  - ▶  $A$ : log price at  $t_c$ ;  $B < 0$  and  $0 < m < 1$ : super-exponential growth;  $C_1, C_2$ : amplitudes of the oscillations;  $\omega$ : their log-frequency
- Two steps [Filimonov and Sornette \(2013\)](#): for given  $(t_c, m, \omega)$  the four linear parameters are OLS; minimise the concentrated sum of squares over three nonlinear parameters
- Qualified fits: the filter of [Shu and Zhu \(2020\)](#) (bounds on  $m, \omega, t_c$ , number of oscillations, damping, relative error, Lomb test, stationary residuals)
- **Confidence indicator**: the share of qualified fits over many windows ending at the same date (here 29 windows of 50–750 days)

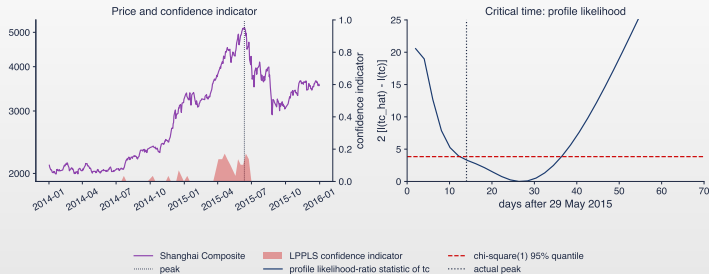
## Inference on the critical time

- The point estimate  $\hat{t}_c$  moves with the window and the data; an **interval** is needed
- **Profile likelihood**: for each  $t_c$ , the other parameters are set to their best values; with Gaussian errors

$$\ell(t_c) = -\frac{n}{2} \ln [\text{SSR}(t_c)/n], \quad 95\% \text{ interval} : \{t_c : 2[\ell(\hat{t}_c) - \ell(t_c)] \leq \chi_{1;0.95}^2 = 3.84\}$$

- ▶  $n$ : number of days in the window;  $\text{SSR}(t_c)$ : the smallest sum of squared residuals for this  $t_c$
- ▶ too narrow when residuals are autocorrelated or when nuisance parameters are many
- [Filimonov, Demos and Sornette \(2017\)](#): the modified profile likelihood (Barndorff-Nielsen) corrects for the nuisance parameters; intervals for  $t_c$  become wider and asymmetric
- [Demos and Sornette \(2017\)](#) ask whether the birth or the burst of a bubble is easier to diagnose: the start time is far better constrained by the data than  $t_c$

## Shanghai 2015: indicator and critical time



- Left: confidence indicator every fifth trading day; right: profile likelihood of  $t_c$  for the window from the low of 20 March 2014 to 29 May 2015 (293 days), as in the real-time analysis of [Sornette et al. \(2015\)](#)

## Interpreting the Shanghai fit

- The fit ending on 29 May 2015 passes the whole filter:  $m = 0.41$ ,  $\omega = 6.28$ , 5.7 oscillations, damping 1.16;  $\hat{t}_c = 24$  June 2015
- Profile-likelihood interval: 12 June 2015 to 2 July 2015 (20 days); the peak of 12 June 2015 lies at its left edge: a good call that is fragile to the window
- The indicator is low: at most 0.17 (on 20 April 2015); few windows qualify at once, so the indicator flags a regime, not a date
- One success is an anecdote: the next section counts alarms over decades, with base rates

## Spectral views of log-periodicity and the critique

- The oscillation  $\cos(\omega \ln(t_c - t))$  is periodic in  $\ln(t_c - t)$ 
  - ▶ test: the Lomb periodogram of the detrended residual against  $\ln(t_c - t)$  [Lomb \(1976\)](#)
  - ▶ nonparametric alternative: the  $(H, q)$ -derivative analysis [Zhou and Sornette \(2003\)](#)
  - ▶ in calendar time its instantaneous frequency is  $\omega/[2\pi(t_c - t)]$ : oscillations speed up towards  $t_c$ ; the Hilbert transform of the residual gives this phase directly ([Chapter 11](#) for the tools)
- Critique
  - ▶ log-periodic peaks appear in noise after fitting and detrending [Feigenbaum \(2001\)](#)
  - ▶ out-of-sample crash calls perform poorly [Brée and Joseph \(2013\)](#); estimation is fragile [Geraskin and Fantazzini \(2013\)](#)
  - ▶ replies on the stochastic formulation, filters and multi-window indicators: [Sornette et al. \(2013\)](#)
- A fair comparison with PSY uses the same events, dates and error measures: next section

## Recap: LPPLS

- LPPLS models super-exponential growth with log-periodic oscillations towards a critical time; the confidence indicator aggregates qualified fits over windows
- The critical time needs an interval: profile or modified profile likelihood; the birth of a bubble is easier to pin down than its burst
- Spectral tests of log-periodicity are easy to fool; judge LPPLS by its alarms, not by its fits

## Five episodes, one protocol



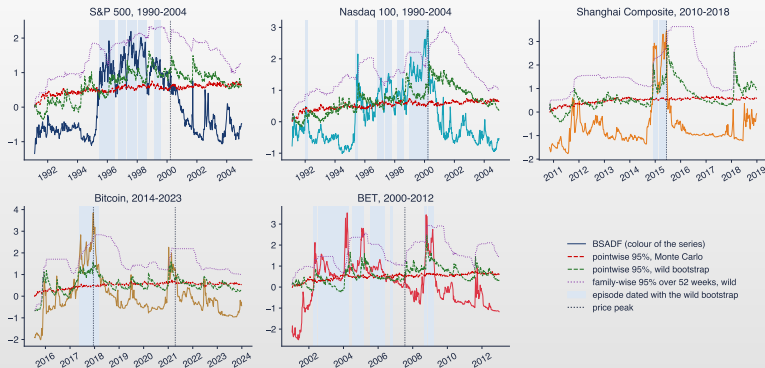
Shanghai Stock Exchange, Pudong, 2008

Photo: Baycrest (2008); CC BY-SA 2.5; Wikimedia

[Commons](#)

- Weekly log prices (Friday close): S&P 500 and Nasdaq 100 1990–2004, Shanghai Composite 2010–2018, Bitcoin September 2014 – 2023, BET 2000–2012
- No lags;  $r_0 = 0.01 + 1.8/\sqrt{T}$ ; minimum duration  $\lceil \ln T \rceil$  weeks; Monte Carlo (1000 paths) and wild-bootstrap (499 draws) critical values
- Episodes dated with pointwise wild values and with the family-wise wild threshold over 52 weeks; the alarm date is the confirmation date
- Case studies: [Phillips, Wu and Yu \(2011\)](#) (Nasdaq), [Phillips, Shi and Yu \(2015a\)](#) (S&P 500), [Sornette et al. \(2015\)](#) (Shanghai), [Gerlach, Demos and Sornette \(2019\)](#) (Bitcoin); for the BET 2007 see also the further reading [Pele and Mazurencu-Marinescu \(2012\)](#)

## BSADF of the five episodes



□ Shaded: episodes dated with the pointwise wild bootstrap; dotted vertical lines: price peaks

## Interpreting the five tests

| Series     | $T$ | GSADF | cv MC | cv wild | episodes: MC / wild / FW | weeks: MC / wild |
|------------|-----|-------|-------|---------|--------------------------|------------------|
| S&P 500    | 782 | 2.19  | 2.28  | 2.91    | 5 / 6 / 0                | 228 / 167        |
| Nasdaq 100 | 782 | 2.92  | 2.28  | 3.22    | 6 / 6 / 1                | 210 / 159        |
| Shanghai   | 460 | 3.46  | 2.15  | 3.90    | 1 / 2 / 1                | 31 / 27          |
| Bitcoin    | 484 | 3.85  | 2.18  | 3.15    | 2 / 2 / 1                | 75 / 43          |
| BET        | 663 | 3.53  | 2.29  | 3.21    | 3 / 7 / 1                | 213 / 217        |

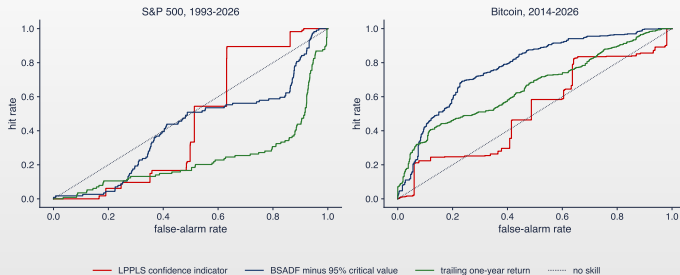
- S&P 500: GSADF below both critical values, yet pointwise dating marks 6 episodes: the multiplicity trap on real data; no family-wise episode
- The wild critical values exceed the Monte Carlo ones for every series (volatility clusters): Bitcoin and BET still reject; the Nasdaq (2.92 against 3.22) and Shanghai (3.46 against 3.90) do not
- MC: Monte Carlo; FW: family-wise wild threshold over 52 weeks; cv: 95% critical value of GSADF

## Alarms before the peaks

| Peak                      | first alarm (2 years before) | lead (weeks) | active at peak | FW lead | fall in 1 year |
|---------------------------|------------------------------|--------------|----------------|---------|----------------|
| S&P 500, 24 March 2000    | 16 April 1999                | 49           | no             | —       | 25%            |
| Nasdaq 100, 24 March 2000 | 24 April 1998                | 100          | yes            | 13      | 65%            |
| Shanghai, 12 June 2015    | 9 January 2015               | 22           | yes            | 22      | 47%            |
| Bitcoin, 15 December 2017 | 23 June 2017                 | 25           | yes            | —       | 82%            |
| Bitcoin, 16 April 2021    | none                         | —            | no             | —       | 49%            |
| BET, 20 July 2007         | 2 September 2005             | 98           | no             | —       | 46%            |

- Every major peak is preceded by a pointwise alarm except Bitcoin 2021; leads of half a year to two years: too early to time a crash, informative about the regime
- BET 2007: the alarm of 2 September 2005 is followed by almost two years of gains before the peak and a fall of 46%; family-wise control removes most alarms
- Counting only peaks hides the false alarms: alarms that were not followed by a crash appear only when every date is evaluated

## Early-warning value over decades



- Event: a fall of 20% within 182 days; evaluation every fifth trading day (S&P 500 1 672 dates, Bitcoin 893); scores: LPPLS confidence indicator, BSADF minus its pointwise 95% critical value (weekly, real time), trailing one-year return

## Interpreting the early-warning evaluation

- Base rates: 6.8% of S&P 500 dates and 47.6% of Bitcoin dates are followed by a 20% fall: the same alarm means very different things in the two markets
- S&P 500, AUC with 90% block-bootstrap intervals: indicator 0.39 [0.32, 0.48], BSADF 0.41 [0.28, 0.58], momentum 0.23 [0.10, 0.40]: all below 0.5
- Most S&P 500 dates followed by a 20% fall lie inside bear markets (2000–2002, 2008), when exuberance scores are low, and few at the top of a run-up
  - ▶ so the scores are **lower** before the events; BSADF alarms have precision 1% against a base rate of 6.8%
- Bitcoin: BSADF has skill, AUC 0.77 [0.69, 0.85], above momentum 0.66; its alarms reach precision 79% against a base rate of 47.6% (hit rate 32%, false alarms 8%)
- The LPPLS indicator is positive on 35.9% of S&P 500 dates and 4.6% of Bitcoin dates; its AUC is 0.39 and 0.49: under this protocol it does not separate pre-crash dates from the rest

## Recap: the applications

- ▣ Wild-bootstrap critical values are higher than Monte Carlo ones on real data; some textbook rejections do not survive
- ▣ Major peaks are usually preceded by alarms, but with long and variable leads; family-wise control removes most of them
- ▣ Over decades, precision must be read against the base rate
  - ▶ exuberance scores help for Bitcoin, where crashes follow run-ups
  - ▶ they fail for the S&P 500, where most pre-crash dates lie inside bear markets

## An open question

- ▣ The question
  - ▶ do explosive-root and LPPLS alarms carry information about large falls beyond momentum and volatility?
  - ▶ once base rates, overlapping horizons and the search over assets are accounted for
- ▣ A testable form
  - ▶ formal: in a logit or probit of the event on lagged momentum, volatility and the alarm score,  $H_0$ : the alarm coefficient is zero, with block-bootstrap inference and a pre-registered panel of assets
  - ▶ falsified by a significant out-of-sample gain in a proper score (log score, Brier) over the momentum-volatility model, pooled across assets
- ▣ Why it matters
  - ▶ regulators and investors read bubble indicators as warnings
  - ▶ most evidence comes from a few famous episodes chosen after the fact
  - ▶ literature to start from: Phillips and Shi (2020), Astill et al. (2018), Demos and Sornette (2017), Brée and Joseph (2013), Homm and Breitung (2012)

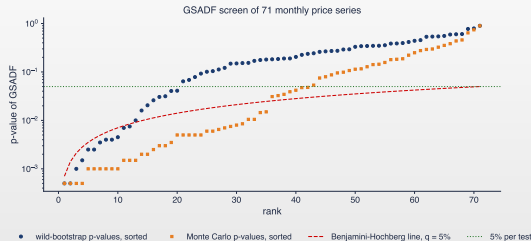
## The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - ▶ **literature:** List peer-reviewed papers since 2015 that evaluate bubble detectors out of sample with false-alarm rates; give DOIs. Then check every DOI and the reported numbers
  - ▶ **hypothesis:** Write a pre-registration: assets, sample, event definition, scores, baseline model, proper score, test and block length.
  - ▶ **code and replication:** ask for a vectorised BSADF; reproduce a known number first (the GSADF of the Nasdaq 100 here, or PSY critical values)
  - ▶ **critique:** Act as a hostile referee: list every way this evaluation could overstate the value of the alarms.
- Report: what was asked, what was kept, what was rejected (AI\_USE.md, AI\_ERRORS.md)

## What the human checks

- ▣ Every reference exists and says what is claimed (DOI resolves, the title matches, the result is in the paper)
- ▣ Right tail, not left: an AI-written ADF often uses left-tail critical values or the 1.645 normal quantile
- ▣ Critical values simulated for the same  $T$ ,  $r_0$ , lags and deterministic terms; wild bootstrap when volatility changes
- ▣ No look-ahead: the minimum window, the lag order and the scaling are fixed from data available at each date
- ▣ Multiplicity over dates and over assets: which error rate does a reported bubble control?

## Mini-case: how many bubbles does a screen find?



- GSADF on the monthly log price of 71 series of the course data (from 1990 or the first month), p-values from 1999 wild-bootstrap and 1999 Monte Carlo draws (smallest attainable p-value: 0.0005)
- Rejections at 5%: 42 with Monte Carlo, 20 with the wild bootstrap (3.6 expected by chance); after Holm 2, after Benjamini-Hochberg 12. An AI summary such as “bubbles in 42 assets” is wrong twice: wrong null and no correction

## Project idea

- **A pre-registered evaluation of bubble alarms on Central and Eastern European markets**
  - ▶ replicate: the date-stamping of [Phillips, Shi and Yu \(2015a\)](#) and the wild bootstrap with family-wise control of [Phillips and Shi \(2020\)](#) on BET, WIG20, BUX, PX and on housing price-to-rent ratios (Eurostat)
  - ▶ extend: compare BSADF, monitoring with a boundary and the LPPLS indicator as predictors of 20% falls, against momentum and volatility, with proper scores and block bootstrap
  - ▶ pre-register: assets, event definition, scores, horizons, tests and the decision rule, before running anything after 1 November 2026
- Deliverables follow the course rules: repository, report, AI\_USE.md, AI\_ERRORS.md, oral defence

## Key takeaways

- ▣ A rational bubble is an explosive autoregressive component; it cannot be negative or start later, and periodic collapses hide it from whole-sample tests
- ▣ Explosive roots are learned at an exponential rate with Cauchy limits; mildly explosive asymptotics are invariant and justify the window tests
- ▣ Critical values must match the design; the wild bootstrap handles changing volatility; date-stamping needs family-wise control
- ▣ Dated starts are late, alarms are early relative to peaks, and precision depends on the base rate
- ▣ Test fundamentals as well as prices; judge LPPLS and PSY by the same alarms, events and scores

## Key formulas

| Object                             | Formula                                                                                                |
|------------------------------------|--------------------------------------------------------------------------------------------------------|
| Bubble                             | $P_t = F_t + B_t, \mathbb{E}_t B_{t+1} = (1 + r)B_t$                                                   |
| Fundamental, random-walk dividends | $F_t = D_t/r + \mu(1 + r)/r^2$                                                                         |
| Explosive OLS limit                | $\rho^n(\hat{\rho} - \rho)/(\rho^2 - 1) \Rightarrow \mathcal{C}$                                       |
| Recursive statistics               | $\text{BSADF}_{r_2} = \sup_{r_1} \text{ADF}'_{r_1}{}^2, \text{GSADF} = \sup_{r_2} \text{BSADF}_{r_2}$  |
| Minimum window                     | $r_0 = 0.01 + 1.8/\sqrt{T}$                                                                            |
| Monitoring boundary                | $b_t = \sqrt{\frac{t-n}{n} \frac{t}{n} [a^2 + \ln \frac{t}{t-n}]}, 1 - \Phi(a) + a\varphi(a) = \alpha$ |
| Precision of an alarm              | $H\pi_0/[H\pi_0 + F(1 - \pi_0)]$                                                                       |
| LPPLS                              | $\ln p = A + Bf + C_1 f \cos(\omega \ln(t_c - t)) + C_2 f \sin(\omega \ln(t_c - t)), f = (t_c - t)^m$  |

## Self-assessment (1/2)

- ▣ Derive  $F_t$  when dividends follow  $D_t = \mu + D_{t-1} + \varepsilon_t$ ; what is  $F_t$  for  $r = 0.04$ ,  $\mu = 0.1$ ,  $D_t = 3$ ?
- ▣ Why can a rational bubble not start after the first trading day?
- ▣ Why does the whole-sample ADF fail against an Evans bubble?
- ▣ Why is the limit of  $\hat{\rho}$  for a fixed explosive root Cauchy only with Gaussian errors?
- ▣ A sample has  $T = 400$  weeks. What are  $r_0$  and the minimum window?
- ▣ Why is the GSADF critical value larger than the SADF one?

## Self-assessment (2/2)

- ▣ What does the wild bootstrap keep from the data, and what does it remove?
- ▣ An alarm has  $H = 0.7$  and  $F = 0.1$ ; the base rate is 4%. What is its precision?
- ▣ Why must the alarm date be the confirmation date and not the dated start?
- ▣ The price-to-rent ratio is explosive and real rents are explosive too. What do you conclude?
- ▣ Why is a profile-likelihood interval for  $t_c$  too narrow with autocorrelated residuals?
- ▣ A screen of 80 assets finds 20 GSADF rejections at 5%. Which two checks come first?

## Self-assessment: answers (1/2)

- ▣  $F_t = D_t/r + \mu(1+r)/r^2 = 75 + 65 = 140$
- ▣ If  $B_t = 0$ , then  $\mathbb{E}_t B_{t+1} = 0$  and  $B_{t+1} \geq 0$  (no negative bubbles), so  $B_{t+1} = 0$  almost surely; by induction the bubble stays at zero
- ▣ Collapses pair the largest falls with the highest levels and pull the OLS slope down; over the whole sample the series looks  $I(1)$  or stationary
- ▣  $\rho^n(\hat{\rho} - \rho)/(\rho^2 - 1) \approx X_n/Y_n$  with  $X_n, Y_n$  dominated by a few shocks: they are normal only if the shocks are normal; no CLT acts inside them
- ▣  $r_0 = 0.01 + 1.8/20 = 0.1$ ; minimum window  $\lfloor 0.1 \cdot 400 \rfloor = 40$  weeks
- ▣ GSADF takes a supremum over more windows (all starts and ends), so under the null its maximum is larger

## Self-assessment: answers (2/2)

- ▣ It keeps the volatility pattern  $|\hat{e}_t|$  (and the short-run dynamics of the null model); random signs remove any explosive or predictable component
- ▣  $0.7 \cdot 0.04 / (0.7 \cdot 0.04 + 0.1 \cdot 0.96) = 0.028 / 0.124 \approx 0.23$
- ▣ The start is known only after the minimum number of exceedances; using it as the alarm date would use future information
- ▣ No evidence of a bubble from these tests: the fundamental explains the acceleration; one would test whether the ratio explodes beyond what rents imply
- ▣ Autocorrelated residuals carry less information than  $n$  independent errors, and nuisance parameters are ignored: the likelihood is too sharp
- ▣ Whether the critical values allow for changing volatility (wild bootstrap); whether the number of rejections survives a multiple-testing correction (4 are expected by chance)

## Further reading and links



The Bucharest Stock Exchange Palace, March  
1928

Photo: Nicolae Ionescu (1928); public domain;  
[Wikimedia Commons](#)

- Surveys
  - ▶ [Gürkaynak \(2008\)](#); [Shiller \(2015\)](#); [Kindleberger and Aliber \(2005\)](#)
  - ▶ bubbles and later returns: [Greenwood, Shleifer and You \(2019\)](#)
- Implementation: the R package `exuber` [Vasilopoulos, Pavlidis and Martínez-García \(2022\)](#)
  - ▶ our Quantlets reproduce its statistics in Python
- Applications to markets, risk and policy: [MFM, Chapter 17](#)
- Further reading on the Bucharest Stock Exchange: [Pele and Mazurencu-Marinescu \(2012\)](#)
- The course ends here; the project defence is described in [Chapter 15](#)

## Appendix: the mildly explosive Cauchy limit

- $\rho_n = 1 + c/k_n$ ;  $X_n = k_n^{-1/2} \sum_{j=1}^n \rho_n^{-(n-j)-1} u_j$ ,  $Y_n = k_n^{-1/2} \sum_{j=1}^n \rho_n^{-j} u_j$ ; weights  $\rho_n^{-j} \approx e^{-cj/k_n}$  decay over  $O(k_n)$  terms
- Lindeberg: no single weight dominates, so  $(X_n, Y_n) \Rightarrow (X, Y)$ , independent  $N(0, \sigma^2/2c)$  ( $\sum_j \rho_n^{-2j} \approx k_n/2c$ ); independence because  $X_n$  and  $Y_n$  load on the two ends of the sample and  $k_n = o(n)$
- $(k_n \rho_n^n)^{-2} \sum_t y_{t-1}^2 \Rightarrow Y^2/2c$  and  $(k_n \rho_n^n)^{-1} \sum_t y_{t-1} u_t \Rightarrow XY$ , so  $\frac{k_n \rho_n^n}{2c} (\hat{\rho} - \rho_n) \Rightarrow X/Y \sim \mathcal{C}$  Phillips and Magdalinos (2007)
- A ratio of two independent centred normals with equal variances is standard Cauchy:  $X/Y = \tan \Theta$  with  $\Theta$  uniform on  $(-\pi/2, \pi/2)$

## Appendix: the monitoring boundary

- Under the null,  $S_t \approx n^{-1/2} \sigma^{-1} [\sum_{j \leq t} \varepsilon_j - (t/n) \sum_{j \leq n} \varepsilon_j]$ ; with  $s = (t - n)/n$ :  
 $S \Rightarrow Z(s) = W(1 + s) - (1 + s)W(1)$ ,  $\text{Var}Z(s) = s(1 + s)$
- Time inversion:  $Z(s) = s B(1 + 1/s)$  for a standard Brownian motion  $B$ ; with  $u = 1/s$  the event  
 $Z(s) > \sqrt{s(1 + s)[a^2 + \ln((1 + s)/s)]}$  becomes  $B(1 + u) > \sqrt{(1 + u)[a^2 + \ln(1 + u)]}$
- Robbins–Siegmund:  $\Pr\{B(1 + u) > \sqrt{(1 + u)[a^2 + \ln(1 + u)]} \text{ for some } u \geq 0\} = 1 - \Phi(a) + a\varphi(a)$ ; the  
 two-sided version doubles it, the form used by [Chu, Stinchcombe and White \(1996\)](#)
- In  $t$ :  $s(1 + s) = \frac{t-n}{n} \cdot \frac{t}{n}$  and  $(1 + s)/s = t/(t - n)$ , which gives  $b_t$ ; a finite horizon makes the test conservative

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